

Introduction to digestion

A rights-clean lecture PDF for PDF-MD extraction testing

- Digestion breaks food into absorbable molecules while moving material through the alimentary canal.
- The sample includes selectable text plus diagrams whose labels are embedded as image pixels.
- Source basis: OpenStax Anatomy and Physiology 2e, Chapter 23, adapted under CC BY 4.0.

System map

Alimentary canal plus accessory organs

- Alimentary canal: mouth, pharynx, esophagus, stomach, small intestine, large intestine, rectum, anus.
- Accessory organs: teeth, tongue, salivary glands, liver, gallbladder, and pancreas.
- The core job is to release nutrients and absorb them into the body while eliminating indigestible residue.

Six digestive processes

Text-table extraction target

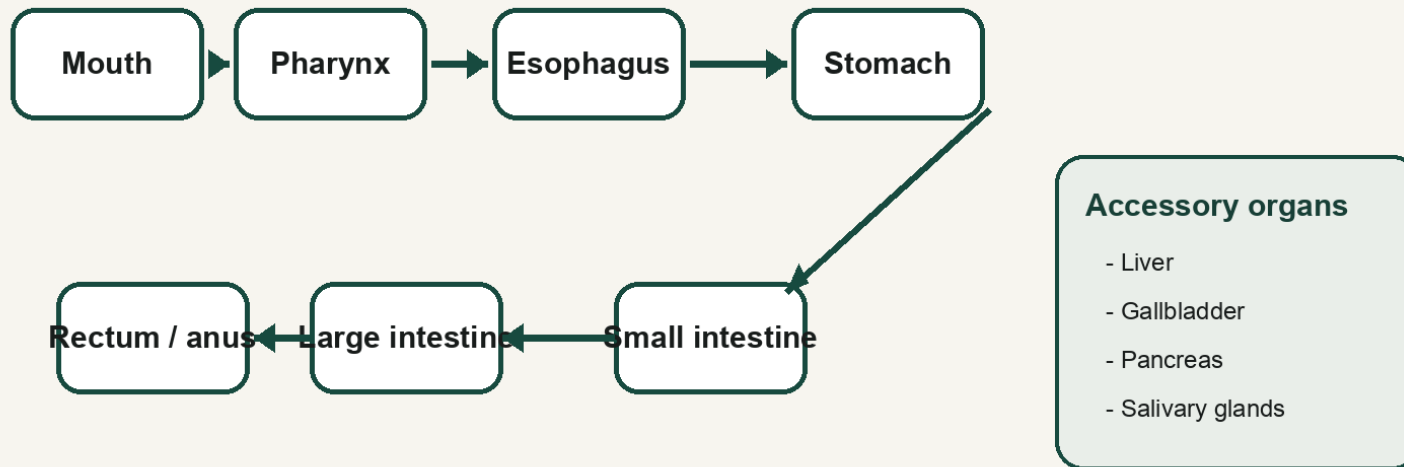
Process	Study cue
Ingestion	Food enters the mouth and is prepared for movement.
Propulsion	Swallowing and peristalsis move contents along the tract.
Mechanical digestion	Chewing, churning, and segmentation increase surface area.
Chemical digestion	Enzymes and acid break macromolecules into absorbable units.
Absorption	Nutrients cross the intestinal mucosa into blood or lymph.
Defecation	Indigestible material leaves the body.

Route map with embedded labels

Image OCR target

Embedded-label route map

The labels in this diagram are pixels, not selectable PDF text.



The diagram labels are part of a PNG, so PDF-MD must use image OCR to recover them.

Normal slide text remains selectable; labels inside the diagram are embedded image text.

Wall layers of the alimentary canal

What students should keep straight

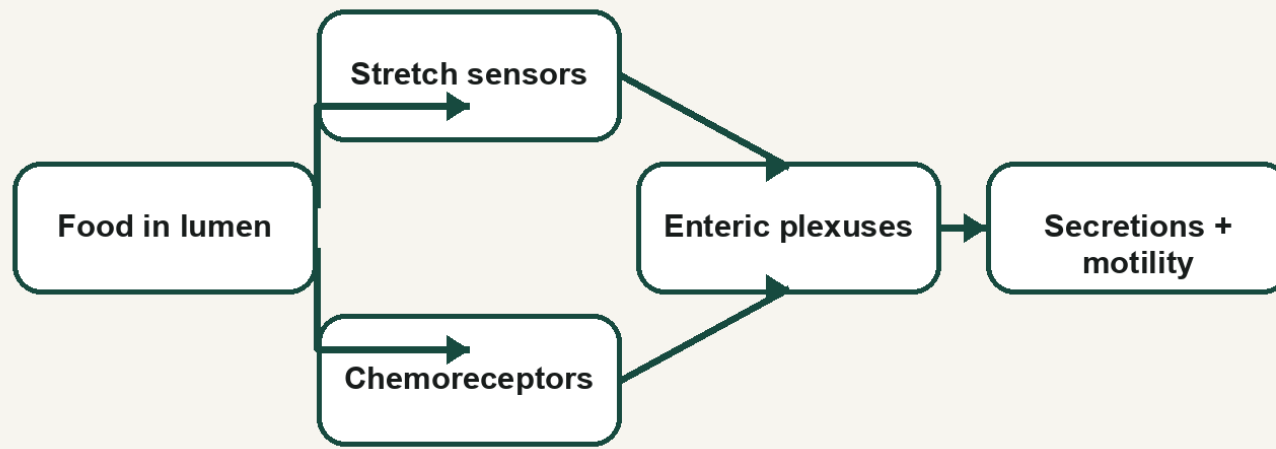
- Mucosa: epithelium and connective tissue at the lumen-facing surface.
- Submucosa: connective tissue, vessels, glands, and local nerve plexus.
- Muscularis: smooth muscle layers that drive mixing and propulsion.
- Serosa or adventitia: outer covering depending on organ position.

Control map with embedded labels

Second image OCR target

Embedded-label control map

Neural and hormonal labels are intentionally embedded in the image.



Study point: regulation is local plus systemic; reflexes and gut hormones tune digestion.

This image tests whether the exported Markdown makes embedded labels visible enough to inspect.

Normal slide text remains selectable; labels inside the diagram are embedded image text.

Why the small intestine matters

Absorption surface area and final digestion

- Most absorption occurs in the small intestine.
- Circular folds, villi, and microvilli increase mucosal surface area.
- Brush border enzymes complete parts of carbohydrate and protein digestion.
- Absorbed nutrients enter blood or lymph depending on the molecule type.

Attribution and sample boundary

Public redistribution note

- Adapted from OpenStax Anatomy and Physiology 2e, Chapter 23: The Digestive System.
- Original OpenStax text is licensed under Creative Commons Attribution 4.0 International.
- This sample is not medical advice. It is a PDF-MD extraction demo for lecture-slide material.
- The test intentionally mixes selectable slide text with embedded text inside diagrams.